

# AirOne — Deployment Guide

This document provides step-by-step instructions for deploying the AirOne Ground Station in various environments.

## Prerequisites

- Python 3.10, 3.11, or 3.12
- 2 GB RAM minimum (4 GB recommended for ML features)
- 500 MB disk space
- USB serial port for CanSat communication (optional for testing)

## Quick Start (Local Development)

*# 1. Clone the repository*

```
cd airone
```

*# 2. Install dependencies*

```
pip install -r requirements.txt
```

*# 3. (Optional) Install ML packages for full detector coverage*

```
pip install scikit-learn torch
```

*# 4. Set JWT secret (required)*

```
export AIRONE_JWT_SECRET="your_secure_secret_minimum_32_characters_long"
```

*# 5. Run the launcher*

```
python3 launcher.py
```

The system will start and be available at: - API: <http://localhost:5000/api/v1/> - GUI: Auto-launches if PyQt5 is available

## Production Deployment

### Option 1: Docker (Recommended)

*# 1. Set your JWT secret*

```
export AIRONE_JWT_SECRET="your_production_secret_32_chars_minimum"
```

*# 2. Build and run*

```
docker-compose up -d
```

*# 3. Check status*

```
docker-compose logs -f airone
```

*# 4. Access API*

```
curl http://localhost:5000/api/v1/system/health
```

## Option 2: Systemd Service (Linux)

Create /etc/systemd/system/airone.service:

### [Unit]

Description=AirOne Ground Station

After=network.target

### [Service]

Type=simple

User=airone

WorkingDirectory=/opt/airone

*# Keep secrets out of the unit file: /etc/airone/env must be root:airone 0640*

EnvironmentFile=/etc/airone/env

*# Default bind is 127.0.0.1 (plain HTTP). Only set AIRONE\_API\_HOST=0.0.0.0*

*# behind a TLS reverse proxy.*

NoNewPrivileges=true

ProtectSystem=strict

ReadWritePaths=/opt/airone/data /opt/airone/logs

ExecStart=/usr/bin/python3 /opt/airone/launcher.py

Restart=on-failure

RestartSec=10

### [Install]

WantedBy=multi-user.target

Bootstrap the first administrator before starting the service:

```
sudo -u airone env $(sudo cat /etc/airone/env | xargs) \  
  AIRONE_ADMIN_PASSWORD='<strong password>' python3 \  
/opt/airone/launcher.py --create-admin
```

Enable and start:

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable airone
```

```
sudo systemctl start airone
```

```
sudo systemctl status airone
```

## Environment Variables

| Variable              | Required   | Default | Description                                                                |
|-----------------------|------------|---------|----------------------------------------------------------------------------|
| AIRONE_JWT_SECRET     | <b>YES</b> | None    | JWT secret (min 32 chars)                                                  |
| AIRONE_ADMIN_PASSWORD | bootstrap  | None    | Password for launcher.py --create-admin / --create-user / --reset-password |
| AIRONE_LINK_KEY       | No         | None    | Hex key (>= 16 bytes) for HMAC                                             |

| Variable                                 | Required | Default     | Description                                                      |
|------------------------------------------|----------|-------------|------------------------------------------------------------------|
| AIRONE_API_HOST                          | No       | 127.0.0.1   | telemetry frame tags<br>API bind address<br>(overrides api.host) |
| AIRONE_SERIAL_PORT                       | No       | Auto-detect | Serial port path                                                 |
| AIRONE_ALLOW_DEFAULT_USERS               | No       | 0           | 1 seeds must-change<br>demo accounts (dev<br>only)               |
| AIRONE_ALLOW_CORS_WILDCARD               | No       | 0           | 1 permits cors_origins:<br>["*"] (dev only)                      |
| AIRONE_GUI_USER /<br>AIRONE_GUI_PASSWORD | No       | None        | GUI login (otherwise<br>GUI is<br>UNAUTHENTICATED<br>)           |
| QT_QPA_PLATFORM                          | No       | Auto        | Set to offscreen for<br>headless                                 |

See .env.example for a template.

## Security Checklist

Before production deployment:

- ☐ Set AIRONE\_JWT\_SECRET to a strong, random 32+ character string (never in YAML)
- ☐ Create the administrator with launcher.py --create-admin (no accounts ship; do **not** set AIRONE\_ALLOW\_DEFAULT\_USERS)
- ☐ Generate a link key and flash it into the firmware (AIRONE\_LINK\_KEY\_HEX); set AIRONE\_LINK\_KEY and, once verified, telemetry.require\_authenticated\_frames: true
- ☐ Keep api.host on 127.0.0.1 unless a TLS reverse proxy sits in front
- ☐ Configure the exact CORS allow-list in config/default\_config.yaml
- ☐ Run pip-audit -r requirements.txt and launcher.py --validate-only
- ☐ Schedule launcher.py --verify-audit and archive logs/audit.jsonl off-host
- ☐ Review RBAC permissions in src/security/rbac.py
- ☐ Enable HTTPS (use a reverse proxy like nginx)
- ☐ Restrict API port access with firewall rules
- ☐ Review audit log rotation settings
- ☐ Ensure serial device permissions are correct

## Serial Device Configuration

### Linux

*# Find your device*

```
ls /dev/ttyUSB* /dev/ttyACM*
```

*# Add user to dialout group*

```
sudo usermod -a -G dialout $USER
```

*# Set environment variable*

```
export AIRONE_SERIAL_PORT=/dev/ttyUSB0
```

### Windows

*# List COM ports*

```
[System.IO.Ports.SerialPort]::getportnames()
```

*# Set environment variable*

```
$env:AIRONE_SERIAL_PORT="COM3"
```

### Docker

Uncomment the devices section in docker-compose.yml:

**devices:**

```
- /dev/ttyUSB0:/dev/ttyUSB0
```

## Database Backup

The system automatically creates online backups, but for manual backup:

*# While system is running*

```
sqlite3 data/airone.db ".backup data/airone_backup_$(date +%Y%m%d_%H%M%S).db"
```

## Troubleshooting Deployment

### “InsecureConfigError: Refusing to start with default JWT secret”

Set a strong AIRONE\_JWT\_SECRET environment variable (min 32 characters).

### “No module named ‘PyQt5’”

The GUI is optional. Either: - Install PyQt5: pip install PyQt5 - Run headless: export QT\_QPA\_PLATFORM=offscreen

### “Serial port not found”

Check: 1. Device is connected: ls /dev/ttyUSB\* 2. User has permissions: groups (should include dialout) 3. Port is not in use: lsof /dev/ttyUSB0

## “Permission denied” on serial port

```
sudo chmod 666 /dev/ttyUSB0
# Or permanently:
sudo usermod -a -G dialout $USER
# Then logout and login again
```

## ML detectors unavailable

Install optional dependencies:

```
pip install scikit-learn torch
```

## Performance Tuning

### For High-Frequency Telemetry (>10 Hz)

Adjust config/default\_config.yaml:

```
ml_worker:
  enabled: true
  window_size: 1000 # Increase buffer
  score_interval: 10 # Score every 10 frames
```

### For Low-Memory Systems (<2 GB RAM)

Disable ML worker:

```
ml_worker:
  enabled: false
```

## Monitoring

### Health Check Endpoint

```
curl http://localhost:5000/api/v1/system/health
```

Returns:

```
{
  "success": true,
  "data": {
    "status": "healthy",
    "workers": {...},
    "database": "connected"
  }
}
```

### Structured Logs

```
# View recent logs
tail -f logs/airone.jsonl | jq .
```

```
# View audit chain
tail -f logs/audit.jsonl | jq .
```

## Upgrading

1. Stop the service
2. Backup the database
3. Pull new code / download new release
4. Check for migration changes
5. Restart the service

### # Docker

docker-compose down

docker-compose pull

docker-compose up -d

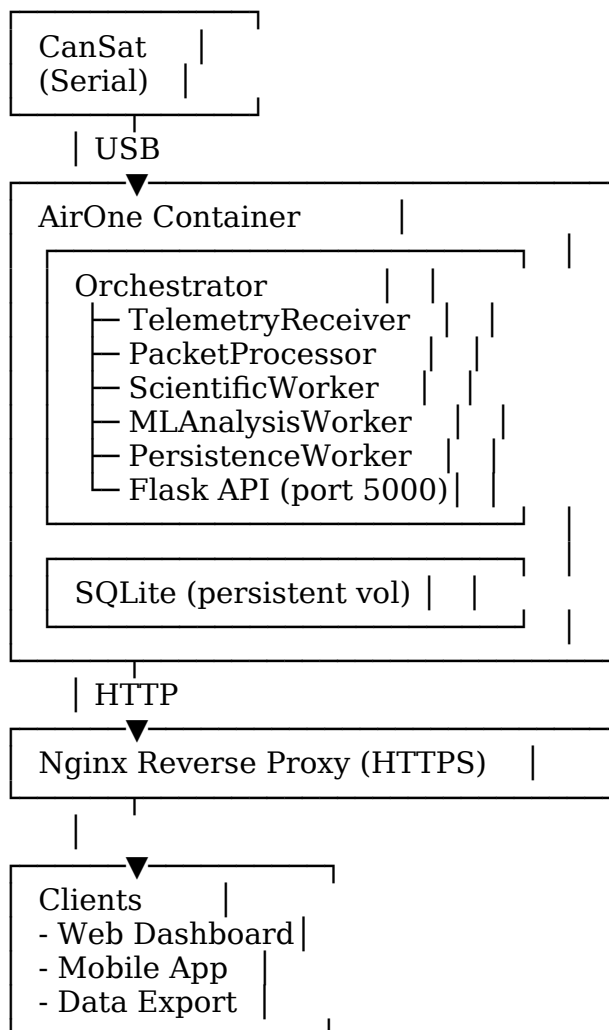
### # Systemd

sudo systemctl stop airone

### # Update code

sudo systemctl start airone

## Production Architecture Example



## Support

For issues, consult: 1. [troubleshooting.md](#) 2. [failure\\_modes.md](#) 3. GitHub Issues (if applicable) 4. System logs in logs/airone.jsonl